AeroSense Bio

FITI (From IP To IPO) Program

ITERATION REPORT

NTHU iGEM 2026

Contents

1. Purpose of This Report 3

2. Timeline Overview 3

3. Stage-by-Stage Iteration Record 4

3.1 Open Call — June 2026 4

3.2 Mentor Session — July 2026 4

3.3 Business Plan Review — August 2026 5

3.4 Pitch Deck Review — August 2026 6

3.5 Supporting-Document Update — 2026-09-09 7

3.6 Evaluation Round One — September 2026 7

4. Three Turning Points That Run Through the Whole Process 9

5. What Is Still Open 9

Appendix: Source Documents 9

1. Purpose of This Report

This report is not a chronological log for its own sake. It is written to make one thing visible: at every stage of the FITI process, the feedback AeroSense Bio received was actually absorbed into a concrete change — to the product narrative, the target market, the competitive framing, or the supporting evidence — rather than being acknowledged and left on the shelf. Each stage below is written in a Design–Build–Test–Learn format: what we assumed going in, what we built or submitted, what the reviewers or mentors told us, and what we changed as a direct result. Read end to end, the sequence also tells a larger story about how the team's own understanding of its business matured over four months, which is summarized separately in Section 4.

2. Timeline Overview

Stage	Date	Format	Feedback Source	Our Response
Open Call	2026-06	Written review	Three FITI reviewers	Informed subsequent revisions; no standalone document produced at this stage
Mentor Session	2026-07	Live discussion, transcribed	Fang Chia-Hung, Su Shih-Chung, Shen Li-Ping, Feng Chih-Feng, Hung Liang-Cheng	Feedback folded into the August revisions below
Business Plan Review	2026-08	Written, 11-point review	Orientation Bootcamp mentors	22-page revised Business Plan
Pitch Deck Review	2026-08	Written, via HackMD	Orientation Bootcamp mentors	18-page revised pitch deck
Supporting-Document Update	2026-09	Internal revision	Consolidated response to the above	v2 of the Creative Work Description
Evaluation Round One	2026-09	Written review	Three FITI reviewers	Ongoing — see Section 5

3. Stage-by-Stage Iteration Record

3.1 Open Call — June 2026

Design (starting assumption). The team entered FITI with its original business framing: a biologically-inspired electronic nose platform, presented primarily on the strength of its underlying sensing technology.

Build. Initial written application materials describing the platform and its intended applications.

Test (what reviewers said). Three reviewers raised distinct concerns. Reviewer 1 questioned what return-on-investment range a target customer could expect, and asked a pointed structural question: after detection, how does the result connect to an actual protective or corrective action — and is the paying customer the end-user directly, or a systems integrator sitting between AeroSense and the end-user? Reviewer 2 found the business plan's research and commercialization strategy clear, and its risk assessment and equity planning well laid out, but flagged that IP risk needed deeper mitigation. Reviewer 3 asked the team to clarify plainly whether its real competition was chemical-sensing technology, biological-sensing technology, or both; asked for more concrete market analysis — the actual scope and scale of existing detection methods in use, the shape of the industry ecosystem, and specific customer problems rather than general statements; noted that although the platform has broad potential, the early stage should stay focused rather than spreading resources across too many applications; and asked for a clearer account of what licensing this venture would require and how that licensing would shape its commercial path.

Learn (what this set in motion). No single document was produced in direct response to the open call, but its three concerns — unclear ROI and customer identity, insufficient IP planning, and a lack of market-specific evidence — became the throughline for nearly everything that followed. Each later stage can be read as a further attempt to answer one or more of these three original questions.

3.2 Mentor Session — July 2026

Design. By this point the team had begun narrowing its story but had not yet built a single, quantified beachhead market or a clear commercialization logic.

Build. A live working session with five mentors: Fang Chia-Hung (a physics graduate of NTHU and founder of the startup Tri-in), Su Shih-Chung (CEO) and Shen Li-Ping (Deputy General Manager), and Feng Chih-Feng (Advisor) and Hung Liang-Cheng (Deputy Director).

Test. The feedback ranged across the full business, and is worth reading as two distinct voices. Su Shih-Chung and Shen Li-Ping focused on go-to-market mechanics and competitive positioning: roughly a year to assemble a working proof-of-concept; NTHU's own incubation track record already includes an electronic-nose company (Enosim Bio-Tech) worth learning from directly, so the team should ask how that company built its market rather than duplicating the effort; a biomedical venture that has to pivot should be able to draw on senior alumni for advice; the team should not assume metal-oxide sensors are automatically at a disadvantage, since there is real revenue to be made in offering calibration services regardless of sensor type; and, most directly, the team should go talk to an actual farmers' association at

an actual agricultural collection point, because an existing solution may already be in place there, and the open question is not whether existing methods have some baseline precision, but whether the customer actually has an unmet need and what price they would accept. Feng Chih-Feng and Hung Liang-Cheng pushed harder on commercial logic: the technology needs to be stated plainly in terms of what industry problem it solves, whether the result is qualitative or quantitative, and how it becomes a sellable product; "another electronic nose" was noted as an unhelpful frame, because it was not clear what advantage or significance the team's approach actually carried; the competitive analysis did not correspond to the hardware-cost problem raised at the outset, the reasoning was not tight enough, and it was unclear who the customer actually is; the team was asked what specifically motivates biological gas detection as opposed to other approaches, why other industries were not considered, and to support the claim of technical readiness with quantitative data and published scientific evidence rather than assertion; on IP, the team was told to look at what the university and research institutes require for licensing and technology transfer, to interview people who actually attend the relevant trade shows, and to build a more concrete sense of who is willing to pay; and, more broadly, the team was reminded that its original motivation — pushing an academic mechanism toward commercial use — has to be judged against a real question: is the cost of the problem being solved (measured, for instance, in actual losses) large enough that a solution is the more economically sound choice. The mentors also suggested a direct conversation with NTHU's own incubation center for further development advice, and noted that the current product page shows scientific data but gives no sense of how that data compares to anyone else's, which makes it land flat for an outside reader.

Learn. No single artifact followed this session directly, but its themes — competitive benchmarking against Enosim Bio-Tech, calibration as a possible revenue line, the instruction to physically go talk to a farmers' association, and the repeated emphasis on IP and licensing — reappear explicitly in the August Business Plan revision and, later, in the team's own decision to commission a formal interview outline for a Enosim Bio-Tech visit.

3.3 Business Plan Review — August 2026

Design. Going into this review, the Business Plan still framed the addressable opportunity as broad "facility agriculture pest and disease early detection," spanning six candidate crops (dragon fruit, passion fruit, strawberry, grape, citrus, and banana).

Build. A formal written review session with Orientation Bootcamp mentors, resulting in eleven numbered points of feedback against the existing Business Plan.

Test. The eleven points, in summary: (1) find the market's actual pain before deciding whether the team's strengths are a fit, not the other way around; (2) visit Enosim Bio-Tech and learn from its ten-year history — the advisors of both teams have collaborated before, so there may be room for real coordination rather than duplicated effort; (3) interview and physically observe real target customers, such as a farmers' association's actual inspection process, to see whether they would genuinely want to replace manual inspection with an electronic-nose device and whether the team's intended price point is realistic — and, for any medical-diagnostic ambitions, the same applies to lab-medicine physicians and their existing testing workflow; (4) confirm, through direct observation, that the customer's need is genuinely unmet rather than already served by an existing solution customers simply have no motivation to replace; (5) if

the industry's real problems are blind spots, bias, spoilage, poor selectivity, high power consumption, and hard-to-detect failure, the proposal should show concretely how it improves on each dimension; (6) use quantitative data wherever possible to state both the industry problem and the venture's own results; (7) although the platform's potential range is wide, the early stage should stay focused rather than spreading resources thin; (8) add the scientific evidence already available to the team; (9) the competitive analysis should map directly onto the industry problems already described, not run on a separate track; (10) add a clear account of current and planned IP positioning; and (11) technical stability and mechanism are still unclear, and the application reads as a direction and an industry rather than an actual product — the team needs to define what it is actually selling as a product or service and estimate a business model around that, since entering FITI itself requires thinking in terms of operations and business model.

Learn. The response was substantial and produced a 22-page revised Business Plan. It was rewritten to commit the first phase to a single beachhead — early banana disease detection — rather than the original six-crop spread. The market-sizing methodology changed entirely, moving from a benchmark against the global electronic-nose instrument market (roughly US$164M growing to US$293M) to a bottom-up agricultural-VOC market estimate tied to actual crop-loss figures (TAM ~NT$113.6B, SAM ~NT$15–18B, SOM ~NT$20M). A new six-point industry-pain-point framework was added and mapped, point by point, to the venture's technical response, with a supporting radar chart. A new chapter formally incorporated the team's existing technical evidence document. The competitive-analysis matrix was rebuilt around the same six pain points rather than a separate set of criteria. A new section addressed IP positioning, distinguishing what the team already holds from what is still to be filed. And Sections 9.1–9.2 defined the actual product specification and pricing for the first time (device ~NT$120,000; cartridge subscription ~NT$4,000/month), sourced directly from the team's own financial model.

3.4 Pitch Deck Review — August 2026

Design. The original 13-page pitch deck presented the venture largely as "we have an electronic-nose technology that can do many things," organized around the same six-crop framing the Business Plan had just moved away from.

Build. A separate, slide-level written review of the pitch deck by Orientation Bootcamp mentors, delivered via HackMD — distinct from the Business Plan review above, though pointed in the same direction.

Test. The reviewers' central line, quoted directly because it became the team's own guiding sentence afterward, appears below. Beyond that framing problem, the reviewers flagged specific weaknesses: the competitive-analysis slide compared detection cycle time, an axis on which the team is not actually competitive, and effectively exposed a weakness rather than a strength; the team slide showed photos and titles without answering why this particular group of people is positioned to execute; the problem statistics were generic, global figures not tied to what a specific shipment of the team's own beachhead crop would actually lose; and the deck lacked pages that showed either a simplified explanation of how the system works or evidence of technical validation already completed.

> *"Right now it reads more like 'we have an electronic-nose technology that can do a lot of things,' but FITI needs to see 'a specific group of customers with a clear, expensive problem — and our odor detection directly reduces their loss, raises their price, or sets a new standard.'"*

Learn. An 18-page revised deck followed. The opening was rebuilt around a concrete scenario — a hypothetical 100-tonne banana export shipment — clearly labeled as illustrative rather than an actual shipment record. The disease and ripeness narrative was consolidated into banana (as two use cases) plus mango (data still pending), with tea explicitly positioned as a second-phase market rather than a co-equal one. New pages were added: How It Works, Technology Validation, the service and payment flow, and a three-phase Go-to-Market plan. The competitive analysis was rebuilt entirely around company-to-company comparison (Alpha MOS, Aryballe, and Sensigent against AeroSense) rather than cycle time. The business model was split into four distinct revenue lines — hardware, consumables, software, and project services. Anything not yet complete — farmers'-association interviews, the patent search, DLIA hardware-validation data, and unit economics — was labeled honestly as "planned" or "in progress" rather than papered over with invented numbers. The team slide was replaced with a capability matrix (Sensor/Hardware, AI/Data, Agriculture/Biology, Business Development) built around the team's existing, real photographs. The venture's final positioning was restated as a B2B Agricultural VOC Intelligence Platform rather than a general-purpose electronic nose, with two near-term beachheads (banana export supply chain; premium tea processing) and a longer-term expansion into food, industrial, environmental, and medical applications.

3.5 Supporting-Document Update — September 2026

Design. As of the August Business Plan revision, several field-evidence items were still honestly labeled "not yet completed" — most notably farmers'-association and facility-farm interviews.

Build. A revision of the Creative Work Description (v2), based on the latest 16-page competition deck submitted separately, incorporating every change from Sections 3.3–3.4 above and reconciling them into a single up-to-date document.

Test / Learn. This stage is itself a piece of evidence rather than a response to new external feedback: it is the moment several previously "planned" items — the Taoyuan District Agricultural Improvement Station visit, local farmers'-association interviews, and facility-farm interviews — were confirmed complete and folded into the document's "supplementary notes" and "development process" sections. The product was formally named "AeroSense 1.0," with its modular VOC Sensor Cartridge design and three deliverable outputs (Disease Risk, Ripeness, Quality Grade) stated explicitly for the first time. Market sizing, competitive positioning, and the four-layer revenue model were brought in line with the revised Business Plan and pitch deck, hardware pricing was revised upward to NT$400,000–500,000 per unit (from an earlier NT$120,000 figure) to reflect a more realistic cost basis, and a three-phase Go-to-Market plan (Pilot → Export Supply Chain → Scale) was formalized. Prof. Ching-Che Charng (China Medical University) was added as a co-advisor alongside Prof. Chung-Chuan Lo, and the core team roster of four members was confirmed with their current academic standing. Items still genuinely unresolved — banana/mango/tea field VOC data accumulation, the formal patent novelty search, and DLIA hardware-validation data — were retained as open items rather than closed out prematurely.

3.6 Evaluation Round One — September 2026

Design. Entering this round, the team's materials reflected every revision above: a single quantified beachhead, a company-to-company competitive framing, an honest evidence-maturity ledger, and a defined product and pricing structure.

Build. A further formal written evaluation by three FITI reviewers of the now-revised materials.

Test. The tone shifted noticeably from the open call. Reviewer 1 found the core concept and core technology clear and genuinely innovative, the differentiation from competitors clearly analyzed, the business model pragmatic — noting explicitly that while the initial entry point is agricultural detection, the core technology can be customized and extended into other industries such as food inspection — and the team's division of labor well organized. Reviewer 2 struck a more cautious note: technical depth is strong and the team's background is excellent, but the business model and on-the-ground need still require hands-on validation; the reviewer asked directly whether the GCaMP6 calcium indicator can actually be used commercially, and whether negotiations over NTHU's technology-transfer equity stake have begun; and raised the business model's central open question plainly — at a device price of NT$120,000 plus a NT$4,000 monthly subscription, will farmers actually make that purchase. Reviewer 3 focused on the gap between capability and outcome: can detection alone translate into a genuine reduction in loss, or is a detection signal alone not yet a full solution; noted that given the characteristics of Taiwan's domestic agricultural sector, the current price point is likely to meet real resistance; and pointed out that agricultural equipment adoption is often entangled with regulatory requirements and government subsidy programs, which the team should understand more deeply as a complementary lever rather than an obstacle.

Learn. This round is still active. Two items raised here — whether GCaMP6 is commercially usable and the status of NTHU's technology-transfer equity negotiation — directly motivated the team's outreach to Prof. Fan Chien-Te, a technology-law faculty member at NTHU, requesting a discussion of exactly these questions. The third reviewer's question about actual farmer buy-in is the same open question the July mentor session and the August Business Plan review both raised independently — it is now the subject of active outreach to several candidate export and packing-house contacts, rather than a question the team is still deferring.

4. Three Turning Points That Run Through the Whole Process

Read across five months, three narrative shifts recur across nearly every stage above, each one first raised as a criticism and then adopted as a deliberate design choice.

From a technology-led story to a customer-pain-led story. The open call asked who the customer actually is and what problem they have. The July mentor session called the deck "another electronic nose" with no stated advantage. The August pitch-deck review gave the team its clearest single line — customers with "a clear, expensive problem," not a technology looking for one. By evaluation round one, reviewers were describing the differentiation as "clear" — the same criticism, resolved.

From six crops to one beachhead. The Business Plan review explicitly named "spreading resources across six candidate crops" as a problem, and the mentor session separately pushed the team to go verify demand at a single, real agricultural collection point rather than assume demand across many. The August revisions narrowed the story to banana as the sole first-phase beachhead, with mango and tea named explicitly as later, not parallel, markets.

From "a cheaper electronic nose" to a platform with a stated economic logic. The earliest materials implicitly competed on being a lower-cost alternative to existing detection technology. Through the Business Plan and pitch-deck revisions, this shifted to a company-to-company competitive comparison and a market size built from actual crop-loss figures rather than instrument-market benchmarks, repositioning AeroSense as a B2B Agricultural VOC Intelligence Platform whose value is stated in avoided loss, not sensor price.

5. What Is Still Open

Three concerns have now been raised independently, by different reviewers, at different stages, without being fully resolved: intellectual-property and technology-transfer positioning (raised at the open call, the mentor session, and evaluation round one); real-world validation that a paying customer will actually adopt the device at its stated price (raised at the mentor session, the Business Plan review, and evaluation round one, each time in slightly different words); and defensible unit economics — actual cost of goods, margin, and a justified pricing logic rather than a stated price alone (raised in the pitch-deck review and implicit in evaluation round one). All three are the direct subject of the team's current outreach — to Prof. Fan Chien-Te; to candidate export companies, packing houses, and cold-chain contacts for the second; and, indirectly, to the hardware mass-production contacts identified in the project's five-stage gap analysis for the third. This report should be treated as current through evaluation round one; the next update to this document should record how those three items were actually resolved, not merely that outreach began.